

NACA 6412 AEROFOIL

Pre-Simulation and Post-Simulation Mathematical Report

Two-Dimensional CFD Study at 16° Angle of Attack

Parameter	Study value
Airfoil	NACA 6412
Chord	1.000 m
Freestream velocity	50.0 m/s
Angle of attack	16°
Flow type	Steady, 2D, turbulent
Solver/model	Segregated flow, k-epsilon turbulence
Domain	50 m × 50 m square; 25 chord lengths in each principal direction
Front / upper / lower boundaries	Velocity inlet, vector (50, 0, 0) m/s
Rear boundary	Pressure outlet, 0 Pa gauge
Aerofoil boundary	No-slip wall
Lift/downforce direction	(0, 1, 0); negative result denotes downforce
Drag direction	(1, 0, 0)
Far-field surface control	Target and minimum surface sizes of 3.0 m on all four outer boundaries
Iterations per design	1,000

Document purpose

This document provides the mathematical basis for the NACA 6412 aerofoil study. Part A records the pre-simulation calculations and expected modelling assumptions. Part B uses the final processed 16° CFD dataset to calculate consistent aerodynamic coefficients, downforce-to-drag performance and post-simulation checks. The complete angle-of-attack study is reported separately in the main portfolio report, where 10° is identified as the maximum-efficiency point, 15° as the recommended balanced high-downforce point and 16° as the maximum-downforce point.

Sign convention. The global positive y-direction is treated as upward. The negative lift result therefore represents aerodynamic downforce, which is the intended output for the inverted Formula 1-style airfoil arrangement.

Known inputs and assumptions

Quantity	Symbol	Value	Unit / note
Chord length	c	1.000	m
Freestream speed	V	50.0	m/s
Density	rho	1.225	kg/m ³ , ISA sea level
Dynamic viscosity	mu	1.789×10^{-5}	Pa·s, ISA sea level
Static pressure	p _{inf}	101,325	Pa, ISA sea level
Speed of sound	a	340.3	m/s, ISA sea level
Angle of attack	alpha	16.0	degrees
Nominal 2D span	b	1.000	m, coefficient convention
Common reference area	S _{ref}	1.000	m ² = chord × nominal span

2D area convention. For a true 2D section, aerodynamic forces are interpreted per metre of span. The nominal area is therefore chord × 1 m. The original 5 m CAD extrusion is not used in the 2D coefficient definition.

Part A: Pre-simulation mathematical report

A1. Aerofoil definition

The NACA 6412 designation defines a four-digit cambered section with 6% maximum camber, the maximum camber located at 40% chord, and 12% maximum thickness. For a 1 m chord:

Geometric feature	Fraction of chord	Dimensional value
Maximum camber	0.06c	0.060 m
Camber position	0.40c	0.400 m from the leading edge
Maximum thickness	0.12c	0.120 m

A2. Angle conversion

$$\alpha_{\text{rad}} = \alpha_{\text{deg}} \times \pi / 180$$

$$\alpha_{\text{rad}} = 16 \times \pi / 180 = 0.2793 \text{ rad}$$

A3. Reynolds number

The Reynolds number compares inertial and viscous effects and is based on the chord length:

$$Re = \rho V c / \mu$$

$$Re = (1.225 \times 50 \times 1.000) / (1.789 \times 10^{-5}) = 3,423,700 \approx 3.42 \times 10^6$$

This Reynolds number should be used when selecting validation data, because lift, drag and separation behaviour depend on Reynolds number.

A4. Mach number and density assumption

$$M = V / a$$

$$M = 50 / 340.3 = 0.147$$

Since $M = 0.147$ is below 0.3, compressibility effects are expected to be limited. A constant-density model is therefore a reasonable first-order assumption for this study.

A5. Dynamic pressure

$$q = 1/2 \rho V^2$$

$$q = 0.5 \times 1.225 \times 50^2 = 1531.25 \text{ Pa} = 1.531 \text{ kPa}$$

A6. Reference area for a 2D section

$$S_{ref} = c \times b$$

$$S_{ref} = 1.000 \times 1.000 = 1.000 \text{ m}^2$$

The same nominal reference area is used for both lift and drag coefficients. This makes the coefficients directly comparable and ensures that the coefficient ratio equals the physical force ratio.

A7. Pre-simulation force equations

$$L = C_L q S_{ref}$$

$$L = C_L \times 1531.25 \times 1.000 = 1531.25 C_L \text{ N per metre span}$$

$$D = C_D q S_{ref}$$

$$D = C_D \times 1531.25 \times 1.000 = 1531.25 C_D \text{ N per metre span}$$

Before the simulation, exact lift and drag cannot be calculated without reliable coefficient values at the same Reynolds number, angle of attack and turbulence/transition condition. The equations above define the expected conversion from CFD coefficients to dimensional force.

A8. Analytical pre-simulation lift prediction

A first-principles lift estimate was produced using thin-aerofoil theory and the NACA 6412 mean-camber line. This is an ideal, inviscid and attached-flow prediction. It is included as a mathematical benchmark rather than an exact prediction at 16° .

$$x/c = (1 - \cos \theta) / 2$$

$$0 \leq \theta \leq \pi$$

$$dy_c/dx = 2m(p - x/c)/p^2 \text{ for } x/c < p$$

$$m = 0.06, p = 0.40$$

$$dy_c/dx = 2m(p - x/c)/(1-p)^2 \text{ for } x/c \geq p$$

$$\text{NACA 6412 mean-camber-line slope}$$

$$\alpha_{L=0} = (1/\pi) \int_0^\pi (dy_c/dx)(1 - \cos \theta) d\theta$$

$$\alpha_{L=0} = -0.10876 \text{ rad} = -6.232^\circ \text{ for the upright section}$$

The inverted section reverses the lift direction. The magnitude of the ideal effective incidence at the simulated orientation is therefore $16^\circ + 6.232^\circ = 22.232^\circ$.

$$|C_{L,ideal}| = 2\pi |\alpha - \alpha_{L=0}|$$

$$|C_{L,ideal}| = 2\pi \times 0.38802 = 2.438$$

$$|L'_{ideal}| = |C_{L,ideal}| q c$$

$$|L'_{ideal}| = 2.438 \times 1531.25 \times 1.000 = 3732.9 \text{ N/m}$$

Interpretation. The ideal pre-simulation prediction is approximately -3.73 kN/m in the global lift direction. Because thin-airfoil theory assumes small incidence and attached inviscid flow, it is expected to overpredict the magnitude at 16°, where viscous separation may be important.

A9. Pre-simulation engineering force envelope

A practical sensitivity envelope was also defined to show how plausible coefficient assumptions translate into force. The assumed ranges are not exact theory or test data. They are transparent design assumptions used to bound the expected CFD result before simulation.

Case	Assumed C _L	Assumed C _D	Predicted downforce	Predicted drag	L /D
Lower loading	-1.60	0.050	2450.0 N/m	76.56 N/m	32.0
Central estimate	-1.80	0.075	2756.25 N/m	114.84 N/m	24.0
Upper loading / drag	-2.00	0.100	3062.50 N/m	153.13 N/m	20.0

Reporting integrity. The ideal thin-airfoil result is the analytical benchmark. The coefficient envelope must be labelled as an assumed sensitivity range, not as a measured or uniquely derived prediction. If these assumptions were not recorded before CFD, the final report should describe them as a retrospective pre-simulation method demonstration.

A10. Supporting pre-simulation quantities

$$p_{0,incompressible} = p_{inf} + q$$

$$p_0 = 101325 + 1531.25 = 102856.25 \text{ Pa}$$

$$\text{Force scaling: } F_2/F_1 = (V_2/V_1)^2$$

At unchanged coefficients, doubling speed produces four times the aerodynamic force.

The incompressible stagnation-pressure estimate provides a useful pressure-scale check. The velocity-squared relationship is valuable for later vehicle-speed sensitivity calculations.

A11. Pre-simulation expectations and acceptance checks

Check	Pre-simulation expectation
Force sign	Negative lift is expected because the inverted section is intended to create downforce.
Compressibility	Limited at M = 0.147; constant density is acceptable for an initial study.
Flow behaviour	At 16°, separation may occur and should be checked using velocity, pressure and wall-shear plots.
Convergence	Residual reduction must be considered together with stable lift and drag monitors.
Reference quantities	$\rho = 1.225 \text{ kg/m}^3$, $V = 50 \text{ m/s}$, $S_{ref} = 1.000 \text{ m}^2$ for both coefficients.
Validation	Reference data should be close to $Re = 3.42 \times 10^6$ and the same airfoil orientation/sign convention.

A12. Confirmed CFD configuration

The final simulation configuration used a 50 m × 50 m two-dimensional domain. The front, upper and lower faces were velocity inlets with a common flow-aligned vector of (50, 0, 0) m/s. The rear face was a 0 Pa gauge pressure outlet and the aerofoil was a no-slip wall. A dedicated surface control applied target and minimum sizes of 3.0 m to all four outer boundaries, maintaining a deliberately coarse far field while local controls resolved the aerofoil, leading edge and wake.

Lift/downforce and drag reports used direction vectors (0, 1, 0) and (1, 0, 0), respectively. Each design was run for 1,000 iterations.

Part B: Post-simulation mathematical report

B1. CFD setup summary

Setting	Implemented model
Dimensionality	Two-dimensional
Time treatment	Steady
Material model	Constant-density air
Flow regime	Turbulent
Turbulence model	k-epsilon
Flow solver	Segregated flow
Inlet direction	Horizontal
Airfoil orientation	Leading edge angled downward by 16°

B2. Raw CFD outputs

Output	Reported value
Lift force	-2735.3 N/m
Drag force	108.6 N/m
Reported lift coefficient	-1.85672
Reported drag coefficient	0.219149
Area associated with x direction	0.3235639 m ²
Area associated with y direction	0.9618438 m ²

B3. Verification of the originally reported coefficients

The reported coefficient values are internally consistent with the two different directional areas entered in the coefficient reports:

$$\begin{aligned}
 D_{\text{check}} &= C_{D,\text{reported}} q A_x \\
 D_{\text{check}} &= 0.219149 \times 1531.25 \times 0.3235639 = 108.6 \text{ N/m} \\
 L_{\text{check}} &= C_{L,\text{reported}} q A_y \\
 L_{\text{check}} &= -1.85672 \times 1531.25 \times 0.9618438 = -2735.3 \text{ N/m}
 \end{aligned}$$

These reproduce the CFD forces closely, showing that the reports were functioning as configured. However, separate projected areas should not be used to compare the lift and drag coefficients of the same 2D aerofoil. A common nominal area is required for a consistent aerodynamic coefficient set.

B4. Corrected lift coefficient

$$C_L = L / (q S_{ref})$$

$$C_L = -2735.3 / (1531.25 \times 1.000) = -1.7863$$

Corrected result. $C_L = -1.7863$. The negative sign identifies downforce under the stated coordinate convention.

B5. Corrected drag coefficient

$$C_D = D / (q S_{ref})$$

$$C_D = 108.6 / (1531.25 \times 1.000) = 0.07091$$

Corrected result. $C_D = 0.07091$. This value uses the same 1 m^2 reference area as the lift coefficient.

B6. Downforce-to-drag ratio

$$|L| / D = |C_L| / C_D$$

$$|L| / D = 2735.3 / 108.6 = 25.19$$

The aerofoil therefore generated approximately 25.2 N of downforce for each 1 N of drag at the simulated operating point.

B7. Equivalent static mass loading

$$m_{equivalent} = |L| / g$$

$$m_{equivalent} = 2735.3 / 9.80665 = 278.92 \text{ kg per metre span}$$

This is only a convenient force comparison. It does not represent an actual increase in vehicle mass.

B8. Summary of final calculated performance

Performance quantity	Final value	Interpretation
Reynolds number	3.424e+06	Operating flow regime
Mach number	0.147	Low-speed, limited compressibility
Dynamic pressure	1531.25 Pa	Freestream kinetic pressure
Downforce	2735.3 N/m	Negative global lift magnitude
Drag	108.6 N/m	Streamwise resistance
Corrected lift coefficient	-1.7863	Common 1 m^2 reference area
Corrected drag coefficient	0.07091	Common 1 m^2 reference area
Downforce-to-drag ratio	25.19	Aerodynamic efficiency at 16°

B9. Pre-simulation versus CFD comparison

Quantity	Analytical / assumed prediction	CFD result	Difference
Downforce: thin-airfoil ideal	3732.9 N/m	2735.3 N/m	26.72% below ideal
Downforce: central estimate	2756.25 N/m	2735.3 N/m	0.76% lower
Drag: central estimate	114.84 N/m	108.6 N/m	5.43% lower
L /D: central estimate	24.00	25.19	4.96% higher

$$\begin{aligned} \text{Difference from prediction (\%)} &= |\text{CFD} - \text{prediction}| / |\text{prediction}| \times 100 \\ \text{Central downforce difference} &= |2735.3 - 2756.25| / 2756.25 \times 100 = 0.76\% \\ \text{Ideal-theory reduction (\%)} &= (3732.9 - 2735.3) / 3732.9 \times 100 \\ \text{CFD downforce is } &26.72\% \text{ below the inviscid thin-aerofoil benchmark} \end{aligned}$$

The difference from ideal theory is physically credible because the theoretical model excludes viscosity, thickness effects and separation. The close agreement with the central engineering estimate should not be treated as independent validation unless that estimate was documented before the CFD result was known.

B10. Circulation implied by the CFD lift

$$\begin{aligned} L' &= \rho V \Gamma \\ \Gamma &= L' / (\rho V) = -2735.3 / (1.225 \times 50) = -44.658 \text{ m}^2/\text{s} \end{aligned}$$

The negative circulation sign follows the selected global force convention and is consistent with an inverted section producing downforce.

B11. Pressure-coefficient relationship for exported surface data

$$\begin{aligned} C_p &= (p - p_{\infty}) / q \\ C_p &= (p - 101325) / 1531.25 \end{aligned}$$

When surface pressure data are exported, this equation can be used to plot upper- and lower-surface pressure coefficient against x/c and to support the physical explanation of downforce and separation.

B12. Interpretation and limitations

The 16° result produces the maximum computed downforce in the investigated sweep, with a downforce-to-drag ratio of 25.19. The negative lift sign is physically appropriate for an inverted automotive aerofoil when positive y is upward.

The 16° case is the maximum-downforce operating point, not the universal optimum. The full study identified 10° as the maximum-efficiency point and 15° as the recommended balanced high-downforce point because 15° retained 99.97% of maximum downforce while reducing drag by 10.47% relative to 16°.

A steady k-epsilon model provides a useful baseline, but 16° incidence may create separated and potentially unsteady flow. Each design was run for 1,000 iterations; however, iteration count alone does not prove convergence. Confidence depends on residual and force-monitor stability, y+, mesh sensitivity and further comparison using a separation-capable model such as k-omega SST or transient URANS.

B13. Report-ready conclusion

Conclusion. At 50 m/s and 16° incidence, the 2D NACA 6412 simulation produced 2735.3 N/m of downforce and 108.6 N/m of drag. Using a consistent 1.000 m² reference area, the corresponding coefficients are $C_L = -1.7863$ and $C_D = 0.07091$, giving a downforce-to-drag ratio of 25.19. The operating Reynolds number is approximately 3.424×10^6 and the Mach number is 0.147. The low Mach number supports the constant-density assumption, while the high incidence requires careful assessment of separation, convergence and turbulence-model sensitivity. Within the complete angle-of-attack study, 16° was the maximum-downforce point; 15° was recommended as the balanced operating point because it retained 99.97% of maximum downforce while reducing drag by 10.47%.

References

1. International Standard Atmosphere sea-level properties: temperature 288.15 K, pressure 101,325 Pa, density 1.225 kg/m³ and speed of sound approximately 340.3 m/s.
2. NASA Glenn Research Center, Reynolds Number: $Re = \rho V L / \mu$ and explanation of inertial-to-viscous similarity.
3. NASA Glenn Research Center, Governing Equations / Drag Equation: aerodynamic force equals coefficient $\times$ dynamic pressure $\times$ reference area.
4. Siemens Community guidance for Simcenter STAR-CCM+: a 2D section is treated with an effective 1 m depth when forming areas.
5. PyAnsys Geometry documentation, NACA four-digit airfoil definition and coordinate equations.